

Group 5

SOA vs. Web Services Architecture:

Understanding Key Differences and ERP Integration

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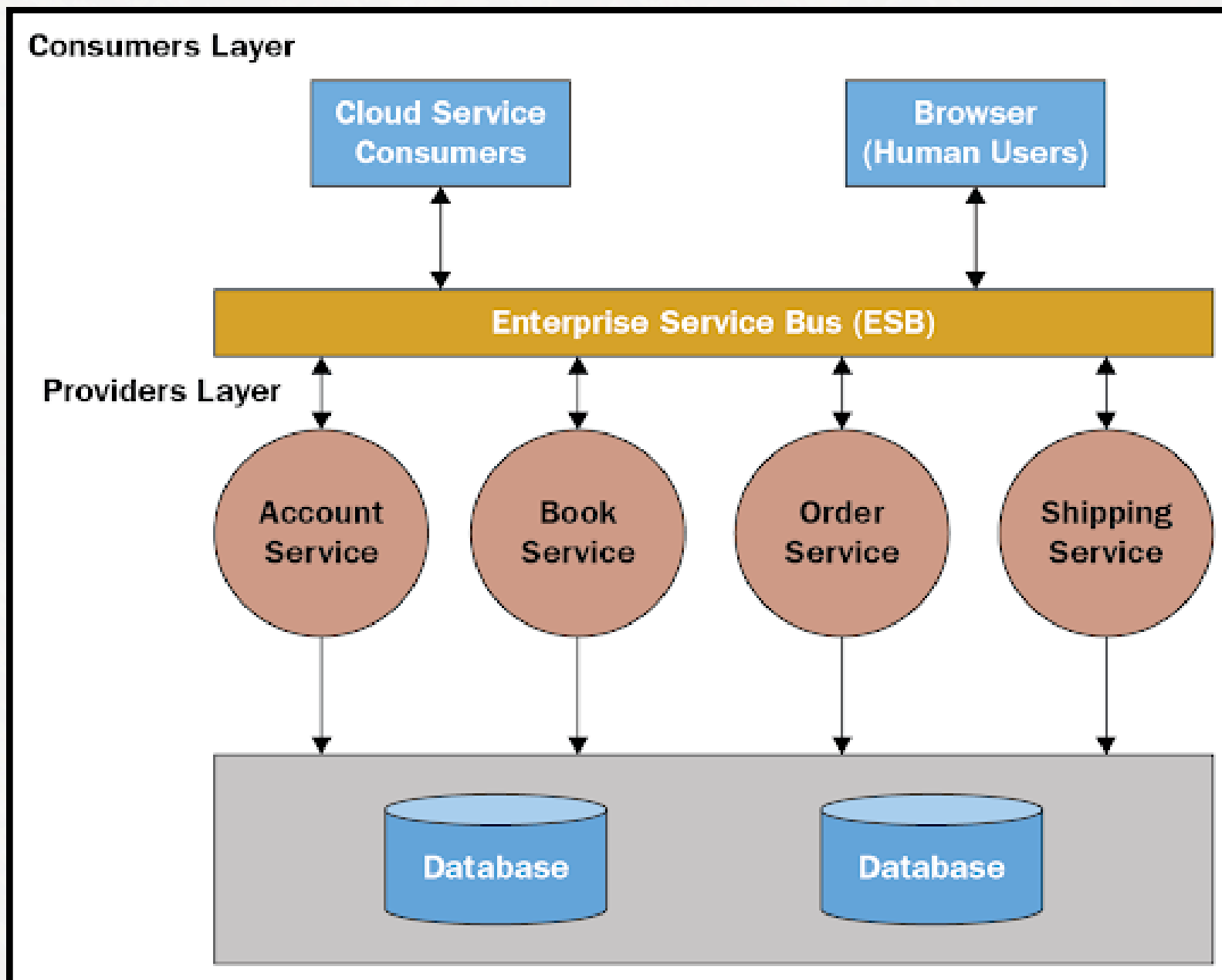
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What is SOA ?



1

A **way of designing systems** where functions are split into **small independent services**.

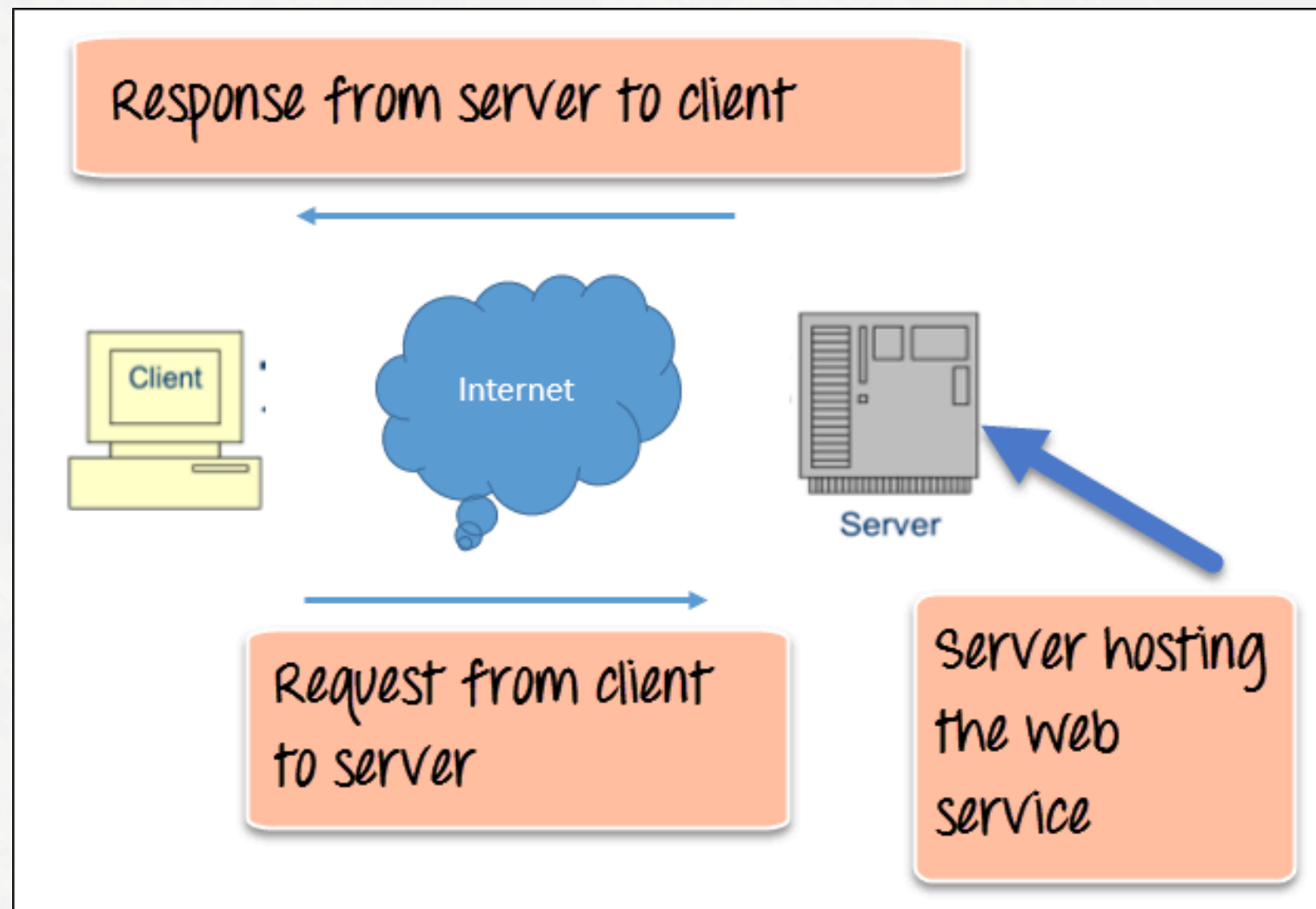
2

Each service does **one job** (like login, payment, or order service).

3

These services can **communicate** with each other through **Enterprise Service Bus (ESB)**.

What is Web Service Architecture ?



- 1 A **technology** used to let two systems talk to each other **over the internet**.
- 2 They use standard formats like **HTTP, XML, or JSON** to send and receive data.
- 3 Example: Grab calls a Google Map API → that's a web service.

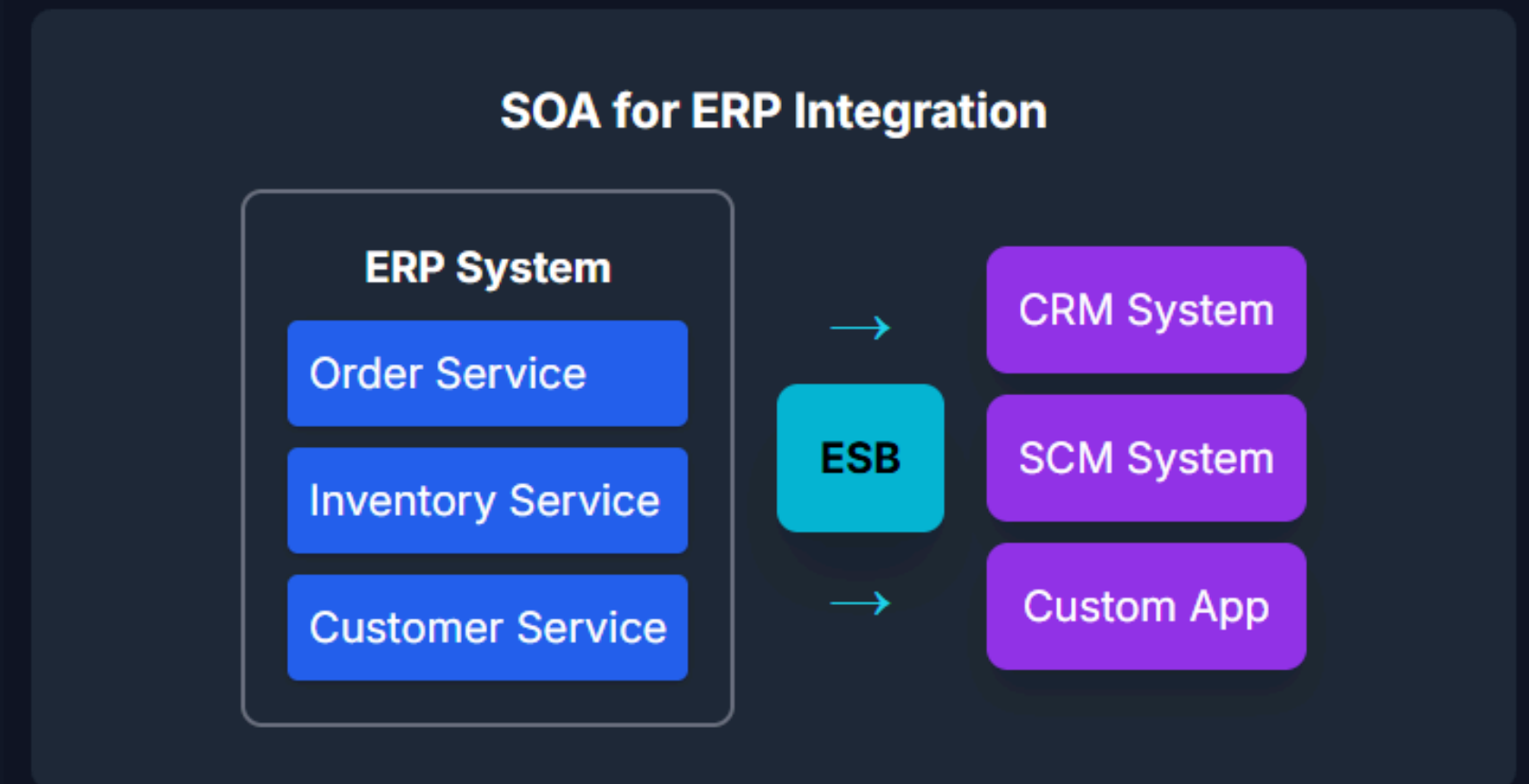
Main Differences: SOA vs. Web Services

Item	SOA (The Plan)	Web Service (The Tool)
What is it?	A set of rules or a big idea for design.	A real technology used to build things.
How big?	Covers the whole company's business.	Connects one program to another.
How does it talk?	Can use many methods (HTTP, etc.).	Almost always uses the web (HTTP).
Standards	Focuses on big ideas (like being "reusable").	Uses specific rules (like SOAP or REST).
Flexibility	Very high (that's the main goal).	Medium (you have to follow its rules).

HOW SOA HELPS ERP

How SOA Helps:

- **Breaks it Down:** Splits the giant ERP into smaller services (like "Check Stock" or "Get Customer Info").
- **Easy to Connect:** These small services are easy for other programs to use.
- **Central Hub (ESB):** Uses a "traffic controller" (ESB) to manage messages between all the services.
- **No Strong Ties:** Other programs (like CRM) can use the ERP services without being stuck to them.



Real Life Example

SERVICE-ORIENTED ARCHITECTURE



Components

1. Student Service – handles student profiles, course enrollment.
2. Finance Service – manages tuition fees, payments, refunds.
3. Library Service – checks book availability, fines, and borrowing records.
4. Hostel Service – manages room assignments and maintenance requests.
5. Service Bus (Enterprise Service Bus - ESB) – connects and routes messages between services.
6. Client Applications – web portals or mobile apps used by students/staff.

WEB SERVICE ARCHITECTURE



Components

1. Service Provider – hosts the web service (e.g. bank server).
2. Service Requester (Client) – user's app or website requesting info.
3. Service Registry – stores available services (account, card, loan).

Conclusion

- SOA is a broad architectural paradigm focused on building flexible, reusable business services.
- Web Services are a specific implementation technology (e.g., SOAP, REST) used to build SOA.
- SOA provides the flexibility and modularity needed to effectively integrate complex systems like ERPs.
- This approach leads to greater business agility and reduced long-term maintenance costs.